

My AoPs Solution Intermediate Algebra

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1 Substitution Exercises

1. Find all ordered pairs (x, y) such that $-x + 2y = 8$ and $x - y = -5$

$$y = x + 5$$

Substituting into the first equation we get $-x + 2(x + 5) = 8$

$$x + 10 = 8$$

$$x = -2$$

Substituting into the second equation we get $2 + 2y = 8$

$$\therefore y = 3$$

$$(x, y) = (-2, 3)$$

2. Find b if a is constant such that $3ax - 8b = 4x + 6a$ for all values of x . Since the equation is true for all values of x it means that x can be set to 0. Thus

$$-8b = 6a$$

It also means that $x = 1$. $\therefore 3a - 8b = 4 + 6a$. Since $-8b = 6a$

$$3a + 6a = 4 + 6a$$

$$3a = 4 \therefore a = \frac{4}{3}$$

Now that we have a we are able to find b .

$$-8b = 6a$$

$$-8b = 8 \therefore b = -1$$

3. Find all ordered pairs (x, y) such that $x + y = 6$ and $x^2 + xy + 2y^2 = 32$. From the first equation we get that $x = 6 - y$
By substitution into the second equation we get

$$(6 - y)^2 + y(6 - y) + 2y^2 = 32$$

$$36 - 12y + y^2 + 6y - y^2 + 2y^2 = 32$$

$$2y^2 - 6y + 4 = 0$$

$$y^2 - 3y + 2 = 0$$

$$(y - 1)(y - 2) = 0$$

$\therefore y = 1, 2$ Then substitute y into the first equation $x + y = 6$ To get that $x = 5, 4$. Therefore the ordered pairs are $(4, 2), (5, 1)$.

4. Solve the system of equations:

$$x + 2y - z = 5$$

$$3x + 2y + z = 11$$

$$(x + 2y)^2 - z^2 = 15$$

Here I can identify the $x + 2y$ in equation 1 and 3. $x + 2y = 5 + z$

$$(5 + z)^2 - z^2 = 15 \quad (1)$$

$$25 + 10z + z^2 - z^2 = 15 \quad (2)$$

$$10z + 25 = 15 \quad (3)$$

$$10z = -10 \quad (4)$$

$$z = -1 \quad (5)$$

Now that we have z we can substitute it into the other equations.

$$x + 2y + 1 = 5 \quad x + 2y = 4 \quad (6)$$

$$3x + 2y - 1 = 11 \quad 3x + 2y = 12 \quad (7)$$

Now the last equation can be rewritten as $2x + (x + 2y) = 12$ $2x + 4 = 12 \therefore x = 4$ Now we substitute x back into the first equation to get $4 + 2y = 4 \therefore y = 0$.

$$\boxed{(x, y, z) = (4, 0, -1)}$$

2 Elimination Exercises

1. Find all ordered pairs (a, b) such that $3a - 2b = -8$ and $5a + 4b = 5$

Multiply the first equation by 2 so that $-2b$ becomes $-4b$, resulting in an easy addition of the two equations.

$$6a - 4b = -16$$

Now add the two equations and we get

$$11a = -11$$

Thus $a = -1$. Substitute back into the first equation to get

$$-3 - 2b = -8 \therefore -2b = -5, b = \frac{5}{2}$$

Thus the ordered pair is $(a, b) = (-1, \frac{5}{2})$.

2. Let x and y be real numbers satisfying $\frac{2}{x} = \frac{y}{3} = \frac{x}{y}$

By cross multiplication between the first and second term we get $6 = xy$, second and third terms result in $y^2 = 3x$
And the first and third terms when cross multiplied result in $2y = x^2$ multiplying both sides by x we get $2xy = x^3$

Since we know that $2xy = 6$ This means that $x^3 = 2(6)$, $\boxed{x^3 = 12}$

3. Find all order pairs (x, y) such that $x^3y^2 = 2^{34}$ and $x^4y = 2^{32}$ Square both sides in the second equation so that we can get y^2 , $x^8y^2 = 2^{64}$ Now we divide the second equation by the first

$$\frac{x^8y^2}{x^3y^2} = \frac{2^{64}}{2^{34}}$$

$$x^5 = 2^{30} \therefore x = 6$$

Substituting this into the first equation we get

$$64^3(y^2) = 2^{34} \therefore y = 256$$

$$\boxed{(x, y) = (64, 256)}$$

4. Find the solution to the system of equations

$$\begin{cases} x + 2z = 1 \\ -xy + 2z = 13 \\ xy + z^2 = 14 \end{cases}$$

I can directly see a similarity between the second and the last equation. Adding the two equations together we can cancel out the xy . This results in

$$\begin{aligned} z^2 + 2z &= -1 \\ z^2 + 2z + 1 &= 0 \\ (z + 1)^2 &= 0 \therefore z = -1 \end{aligned}$$

Substituting z into the first equation we get $x - 2 = 1 \therefore x = 3$ Substitute x and z into equation two resulting in $-3y - 2 = 13$

$$-3y = 15 \therefore y = -5$$

$$\boxed{(x, y, z) = (3, -5, -1)}$$

5. Find all r and s such that $r^2 - 2rs = 27$ and $rs - 2s^2 = 9$ The equations can be rewritten as $r(r - 2s) = 27$ and $s(r - 2s) = 9$. Divide the equations to get rid of the factors

$$\begin{aligned} \frac{r(r - 2s)}{s(r - 2s)} &= \frac{27}{9} \\ \frac{r}{s} &= 3 \therefore r = 3s \end{aligned}$$

Substitute into the second equation to get $3s^2 - 2s^2 = 9$

$$s^2 = 9 \therefore s = \pm 3$$

When $s = 3, r = 9$ and when $s = -3, r = -9$

$$\boxed{(r, s) = (9, 3), (-9, -3)}$$

3 Larger Systems of Linear Equations

1. Solve each of the three systems of linear equations.

$$(a) \begin{cases} 2a + b + c = 2 \\ 5a - b + 3c = 5 \\ 7a + 4c = 8 \end{cases}$$

Adding the first two equations we get $7a + 4c = 7$ which causes the linear system to be inconsistent. No Solution

$$(b) \begin{cases} 2a + b + c = 3 \\ 5a - b + 2c = 5 \\ 7a + 4c = 8 \end{cases}$$

Adding the first equation to the second equation we get $7a + 4c = 8$.

Which means that equation (3) is a redundant equation

That means that system has infinitely many solutions.

Solving for c we get

$$c = \frac{8 - 7a}{4}$$

Substituting into the first equation we get

$$2a + b + \frac{8 - 7a}{4} = 3$$

$$8a + 4b + 8 - 7a = 12$$

$$4b + a = 12$$

$$b = \frac{12 - a}{4}$$

Now we have to solve for c in terms of a . From the last equation we can see that $4c = 8 - 7a$

$$c = \frac{8 - 7a}{4}$$

Thus $(a, b, c) = \left(a, \frac{12 - a}{4}, \frac{8 - 7a}{4}\right)$

$$(c) \begin{cases} x + y + z = 13 \\ x + y - z = -1 \\ x - y + z = 9 \end{cases}$$

By inspection the second and third equations can be added together to remove 2 variables doing so results in $2x = 8 \therefore x = 4$ Adding equation 1 and equation 3 we get

$$2x + 2z = 22$$

$$x + z = 11$$

Now substituting $x + z = 11$ into the first equation we get $y + 11 = 13 \therefore y = 2$

Now substitute all of these into equation one to get $4 + 2 + z = 13$

$$z = 7$$

$$(x, y, z) = (4, 2, 7)$$

2. Given linear system $\begin{cases} a + b + c = 6 \\ a + 2b + 3c = 14 \\ 2a + 5b + 8c = 36 \end{cases}$

Find constants k_1 and k_2 such that k_1 times the first equation plus k_2 times the second equation equals the third equation.

Now the way I solved it was a brute force and not what was intended. I noticed that $3 * 14 - 6 = 36$ This can be seen as $k_1 = -1$ and $k_2 = 3$

The actual way to solve it, which I had considered before finding this out was to use algebra and create a linear system relating k_1 and k_2 to the coefficients of (a, b, c) This is done by $k_1(a + b + c) + k_2(a + 2b + 3c) = 2a + 5b + 8c$. Here we would then create a linear system relating the k and the coefficients of the third equation

3. Solve each of the following systems of equations

$$(a) \begin{cases} a + 3b - 2c = 18 \\ 2a - 4b + 2c - d = -16 \\ -a + 2b - 5c + d = 23 \\ 3b - 7c + d = 35 \end{cases}$$

Subtract the third equation from the last equation. This results in $a + b - 2c = 12$. Now from the first equation we can see that $-2c = 18 - a - 3b$ Substitute this back we get

$$\begin{aligned} a + b - 1 - 3b + 18 &= 12 \\ -2b &= -6 \\ b &= 3 \end{aligned}$$

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 1 \\ x_1 + x_2 - x_3 = 2 \\ x_2 + x_3 = 0 \end{cases}$$

From the last equation we can see that $x_2 = -x_3$ Now substitute this into all of the equations we get

$$\begin{cases} x_1 + x_4 = 1 \\ x_1 - 2x_3 = 2 \end{cases}$$

- (8) This results in 3 variables and 2 equations. This means
 (9) that the system has infinite solutions. We can solve for x_1
 (10) in terms of x_3 and x_4 . From the second equation we can see
 that $x_1 = 2 + 2x_3$. Substitute this into the first equation to
 get $2 + 2x_3 + x_4 = 1$

Now I have trouble working out when there are more than 3 variables, so I rewrite the system.

$$\begin{cases} a - 2c = 9 \\ 2a + 2c - d = -4 \\ -a - 5c + d = 17 \\ -7c + d = 26 \end{cases}$$

$$x_4 = -1 - 2x_3$$

Now we have all of the variables in terms of x_3

$$(x_1, x_2, x_3, x_4) = (2 + 2x_3, -x_3, x_3, -1 - 2x_3)$$

Add the third and the second equation to get $a - 3c = 13$.
 Now from the first equation we can see that $a = 9 + 2c$.
 Substitute this to get $9 + 2c - 3c = 13$

$$-c = 4, c = -4$$

Now substitute c into the last equation to get

$$\begin{aligned} 28 + d &= 26 \\ d &= -2 \end{aligned}$$

$$(a, b, c, d) = (1, 3, -4, -2)$$

4. Suppose we have a system of 3 equations in 3 variables. Such that a times the first equation plus b times the second equation equals the third equation. Where a and b are non zero constants.

(a) Must there be constants c and d such that c times the second equation plus d times the third equation equals the first equation?

Well from the information given $a(eq_1) + b(eq_2) = eq_3$. Now solving for eq_1 we get $eq_1 = \frac{1}{a}eq_3 - \frac{b}{a}eq_2$. This means that $c = -\frac{b}{a}$ and $d = \frac{1}{a}$. (b) If the answer to part (a) is "yes", is it possible to find c and d in terms of a and b ? Is we

can find c and d in terms of a and b as shown above.

$$c = -\frac{b}{a}, d = \frac{1}{a}$$